

TRACE-FREE $SU(2)$ CHARACTERS AND $\mathbb{Z}/4$ INSTANTON GRADINGS FOR TWO-BRIDGE AND $(3, n)$ -TORUS KNOTS

BERND J. WUEBBEN

ABSTRACT. We assemble, and where possible independently verify, the representation-theoretic data underlying the pillowcase (symplectic) side of the Atiyah–Floer conjecture for knots, for two families: the two-bridge knots and the $(3, n)$ -torus knots. For a two-bridge knot $\mathfrak{b}(p, q)$ we give a short, self-contained proof that every irreducible trace-free $SU(2)$ representation is binary-dihedral; these are the $(p-1)/2$ dihedral characters at meridian angles $\cos(2\pi k/p)$, independent of q , and the trace-free Riley polynomial is the explicit product $\phi_p(u) = \prod_k (u + 4 \sin^2(\pi k/p))$, monic of degree $(p-1)/2$ with constant term $\det K$. We use this to give a transparent account of the Hedden–Herald–Kirk theorem that pillowcase homology equals reduced singular instanton knot homology $I^\natural$ on this family, and of why the figure-eight bubbling and bounding cochains that obstruct the general conjecture are structurally inert there. For the $(3, n)$ -torus knots we compute the full trace-free character variety and prove a dihedral dichotomy: exactly $(\det -1)/2$ characters are dihedral, so for n odd every irreducible trace-free character is *non*-dihedral. Passing to the double branched cover $\Sigma(2, 3, n)$, we self-compute the $\mathbb{Z}/4$ spectral-flow gradings of the generators from the Fintushel–Stern index and the equivariant ρ -invariant, calibrated against the Poudel–Saveliev and Anvari computations; for n odd the gradings split evenly between 1 and 3, giving the singular-instanton *chain* complex $IC^\natural(T(3, n)) = (1 + a, a, a, a)$, $a = -\sigma/4$. The *homology* $I^\natural$ equals this for $n \equiv 1 \pmod{6}$ (differential zero) but is smaller by 2 for $n \equiv 5$, where the differential is nonzero: $\text{rank } I^\natural = \sum_i |\Delta_{T(3, n)}|$ throughout, and $T(3, 5) = P(-2, 3, 5) = 10_{124}$ has rank 7, not 9. Finally we reproduce the first nonzero pillowcase differential, for $8_{19} = T(3, 4)$, and identify it as the corner figure-eight bigon that is absent on two-bridge knots. Throughout we distinguish carefully between results proved here, classical facts, and computations reproduced from the literature.

1. INTRODUCTION

1.1. The Atiyah–Floer conjecture and its knot version. Let Y be a closed oriented 3-manifold with a Heegaard splitting $Y = Y_0 \cup_\Sigma Y_1$. Two Floer theories attach to this picture. On the gauge-theoretic side is Floer’s instanton homology $HF^{\text{inst}}(Y)$, generated by flat $SU(2)$ connections and with a differential counting anti-self-dual instantons on $Y \times \mathbb{R}$. On the symplectic side, the moduli space $M(\Sigma)$ of flat $SU(2)$ connections on Σ is symplectic, the handlebodies determine Lagrangians $L_0, L_1 \subset M(\Sigma)$, and one forms the Lagrangian Floer homology $HF^{\text{symp}}(L_0, L_1)$. The Atiyah–Floer conjecture [2] asserts these are isomorphic. The one general theorem is the mapping-torus case of Dostoglou–Salamon [5]; the Heegaard-splitting statement remains open, obstructed by the singularities of $M(\Sigma)$ and by figure-eight bubbling in the adiabatic limit.

There is a knot-theoretic analogue in which the pillowcase plays the role of $M(\Sigma)$. Let $K \subset S^3$ be a knot, and cut S^3 along a Conway sphere meeting K in four points into two 2-string tangles. A representation $\rho: \pi_1(S^3 \setminus K) \rightarrow SU(2)$ is *trace-free* if every meridian maps to a trace-free element (conjugate to $\text{diag}(i, -i)$); this is the boundary condition of Kronheimer–Mrowka’s singular instanton theory [11, 12]. Restricting trace-free flat connections of each tangle to the four-punctured sphere and applying holonomy perturbations, one obtains immersed Lagrangians

L_0, L_1 in the *pillowcase*

$$P = (\mathbb{R}/2\pi\mathbb{Z})^2/\iota, \quad \iota(\gamma, \theta) = (-\gamma, -\theta),$$

the traceless $\mathrm{SU}(2)$ character variety of the four-punctured sphere: a two-sphere with four $\mathbb{Z}/2$ orbifold points [13, 7]. The *knot Atiyah–Floer conjecture* of Hedden–Herald–Kirk and Cazassus–Herald–Kirk–Kotelskiy [7, 3] asserts that a bounding-cochain-deformed Lagrangian Floer homology of (L_0, L_1) is isomorphic, as a relatively $\mathbb{Z}/4$ -graded group, to the reduced singular instanton knot homology $I^\natural(K)$.

1.2. Purpose and scope. This paper is a computational and expository consolidation. Its aim is to make explicit, for two tractable infinite families, the representation-theoretic data on which the pillowcase construction rests: the trace-free character variety (the generators), the $\mathbb{Z}/4$ spectral-flow gradings, and the differential. We are careful throughout to separate three kinds of statement:

- (i) results proved here (self-contained, though some are surely classical);
- (ii) classical facts and standing theorems (attributed);
- (iii) computations reproduced from [8, 14, 1] (attributed, and independently checked where possible).

We do *not* claim any new theorem about the knot Atiyah–Floer conjecture itself: for two-bridge knots it is a theorem of Hedden–Herald–Kirk [7, 8], and for the torus knots we treat the instanton side and the character-variety side, not the analytic identification of the two. The explicit verifications reported below—the exact-arithmetic check of Theorem 1.1, the counts of Proposition 4.1, and the gradings of Section 5—are carried out by a short computer computation included with the source.

1.3. Results. Section 2 records the pillowcase and the trace-free characters as generators. Section 3 proves:

Theorem 1.1 (Dihedral rigidity for two-bridge knots). *Let $K = \mathfrak{b}(p, q)$ be a two-bridge knot, $p = \det K$. Every irreducible trace-free $\mathrm{SU}(2)$ representation of $\pi_1(S^3 \setminus K)$ is binary-dihedral. There are exactly $(p-1)/2$ of them, at meridian-pair angles $\cos \omega = \cos(2\pi k/p)$ for $k = 1, \dots, (p-1)/2$, independent of q ; equivalently the trace-free Riley polynomial in $u = 2 \cos \omega - 2$ is*

$$\phi_p(u) = \prod_{k=1}^{(p-1)/2} \left(u + 4 \sin^2 \frac{\pi k}{p} \right),$$

monic of degree $(p-1)/2$, with integer coefficients, $\phi_p(0) = p = \det K$, and root-sum $-p$.

We then explain (Section 3.2) how Theorem 1.1 clarifies the Hedden–Herald–Kirk two-bridge isomorphism and localizes the obstruction, and we correct a naive expectation about the earring correspondence.

Section 4 computes the $(3, n)$ -torus knot characters and proves:

Theorem 1.2 (Dihedral dichotomy for $(3, n)$ -torus knots). *Let $\gcd(3, n) = 1$. The irreducible trace-free $\mathrm{SU}(2)$ characters of $T(3, n)$ are in bijection with the admissible representation arcs of Section 4, one per arc. Exactly $(\det - 1)/2$ of them are binary-dihedral: one if n is even ($\det = 3$) and none if n is odd ($\det = 1$). In particular, for n odd every irreducible trace-free character of $T(3, n)$ is non-dihedral.*

Section 5 passes to the double branched cover $\Sigma(2, 3, n)$ and self-computes the $\mathbb{Z}/4$ gradings, obtaining for n odd the chain complex $IC^\natural(T(3, n)) = (1 + a, a, a, a)$, $a = -\sigma/4$, whose homology $I^\natural$ equals it for $n \equiv 1 \pmod{6}$ but is smaller by 2 for $n \equiv 5 \pmod{6}$ (where the differential is nonzero),

the even $\mu \in \{1, 3\}$ split accounting for the chain (Proposition 5.2). Section 6 reproduces the 8_{19} differential and identifies its geometric source. Section 7 states the genuinely open direction.

Acknowledgements. The literature computations reproduced here are due to Hedden–Herald–Kirk [7, 8], Cazassus–Herald–Kirk–Kotelskiy [3], Poudel–Saveliev [14], and Anvari [1]; we have tried to attribute each precisely.

2. THE PILLOWCASE AND TRACE-FREE CHARACTERS

We work with $SU(2)$ as the unit quaternions. An element is trace-free iff it is a purely imaginary unit quaternion (eigenvalues $\pm i$, squaring to -1); as a rotation it is a rotation by π . The trace-free elements form a 2-sphere $S \subset \text{Im } \mathbb{H} \cong \mathbb{R}^3$.

Proposition 2.1 (Classical; cf. [13, 7]). *The traceless $SU(2)$ character variety of the four-punctured sphere is the pillowcase $P = (\mathbb{R}/2\pi\mathbb{Z})^2/\iota$, with four $\mathbb{Z}/2$ orbifold points at $(\gamma, \theta) \in \{0, \pi\}^2$ and the symplectic form descending from $d\gamma \wedge d\theta$.*

Sketch. Write π_1 of the four-punctured sphere as $\langle X_1, \dots, X_4 \mid X_1 X_2 X_3 X_4 = 1 \rangle$ with each $X_i \in S$. Since purely imaginary units satisfy $X_i^{-1} = -X_i$, the relation gives $X_1 X_2 = X_4 X_3 =: A$. For pure imaginary units, $X_i X_j = -\langle X_i, X_j \rangle + X_i \times X_j$, so when $A \neq \pm 1$ its axis is well defined; equating the two expressions for A forces the four X_i to be coplanar, orthogonal to $\text{axis}(A)$. Placing that axis along k and writing $X_m = \cos \beta_m i + \sin \beta_m j$, the relation becomes $\beta_2 - \beta_1 = \beta_3 - \beta_4$; gauge-fixing $\beta_1 = 0$ and setting $\gamma = \beta_2$, $\theta = \beta_4$ leaves two parameters, and the residual conjugation by the π -rotation about i acts by $(\gamma, \theta) \mapsto (-\gamma, -\theta)$. The four fixed points are the reducible (collinear) characters. $\square$

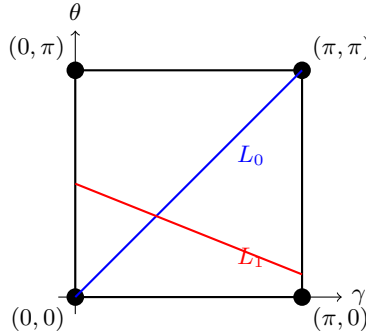


FIGURE 1. The pillowcase $P = (\mathbb{R}/2\pi\mathbb{Z})^2/\iota$, a 2-sphere with four $\mathbb{Z}/2$ orbifold (corner) points, shown on the square $[0, \pi]^2$. For a two-bridge knot the two tangle Lagrangians are straight arcs of rational slope (Section 3.2); being straight they meet transversally and bound no bigon. The earring correspondence bends such an arc into a figure-eight at each corner (Remark 3.2).

For a knot K with a two-tangle decomposition, a representation of $\pi_1(S^3 \setminus K)$ is a compatible pair of tangle representations, i.e. a point of $L_0 \cap L_1$. Thus the intersection points are exactly the trace-free representations of the knot group, and (after perturbation) the generators of the singular instanton chain complex [11, 12]. In what follows a representation is determined up to conjugacy by the single invariant $\cos \omega = \langle a, b \rangle$, the cosine of the angle between the axes of two chosen meridians $a, b \in S$.

3. TWO-BRIDGE KNOTS

3.1. Trace-free characters are dihedral. Recall Riley's parametrization [15]: for a two-bridge knot the nonabelian $\mathrm{SL}_2(\mathbb{C})$ representations with meridian eigenvalue s are

$$\rho(a) = \begin{pmatrix} s & 1 \\ 0 & s^{-1} \end{pmatrix}, \quad \rho(b) = \begin{pmatrix} s & 0 \\ -u & s^{-1} \end{pmatrix},$$

cut out by a polynomial $\Phi_{p,q}(s, u)$ of degree $(p-1)/2$ in u ; the trace-free locus is $s = i$.

Proof of Theorem 1.1. A binary-dihedral representation sends meridians into the reflection coset $e^{i\phi}j$ of $\mathrm{Pin}(2) = S^1 \cup jS^1$, whose elements are trace-free; hence every irreducible dihedral representation is trace-free, so there are *at least* $(p-1)/2$ irreducible trace-free characters (the number of irreducible metabelian representations of a knot with $|H_1(\Sigma_2)| = p$ odd). On the other hand the trace-free Riley polynomial $\Phi_{p,q}(i, u)$ has u -degree at most $(p-1)/2$, so there are *at most* $(p-1)/2$. The bounds coincide, so every irreducible trace-free character is one of the dihedral ones. A dihedral character with $\rho(ab)$ of rotation angle $4\pi k/p$ has $\cos \omega = \cos(2\pi k/p)$; distinct $k \in \{1, \dots, (p-1)/2\}$ give distinct characters, and these values depend only on p . Finally $\phi_p(u) = \prod_k (u - u_k)$ with $u_k = 2 \cos(2\pi k/p) - 2 = -4 \sin^2(\pi k/p)$; its constant term is $\prod_k 4 \sin^2(\pi k/p) = p$ and its root-sum is $\sum_k (2 \cos(2\pi k/p) - 2) = -1 - (p-1) = -p$. $\square$

Remark 3.1. Small cases: $\phi_3 = u + 3$, $\phi_5 = u^2 + 5u + 5$, $\phi_7 = u^3 + 7u^2 + 14u + 7$, $\phi_9 = u^4 + 9u^3 + 27u^2 + 30u + 9$. For p prime, ϕ_p is, after $u = 2c - 2$, the minimal polynomial of $2 \cos(2\pi/p)$, irreducible of degree $(p-1)/2$. Theorem 1.1 is elementary and is essentially classical (the trace-free/metabelian coincidence for two-bridge knots is implicit in [15]); we include the proof because it fixes the interior generators exactly and q -independently, which is what the pillowcase picture needs. The statement was verified by exact Gaussian-integer computation of the Riley word for all $p \leq 9$, including the two determinant-9 knots $9_1 = \mathfrak{b}(9, 1)$ and the hyperbolic $6_1 = \mathfrak{b}(9, 2)$, which produce the identical root set $\{-4 \sin^2(\pi k/9)\}$.

3.2. The pillowcase consequence and the earring. The character variety of a rational tangle of slope r is a straight arc of slope r in the pillowcase, and the double branched cover of the four-punctured sphere is a torus T^2 on which the tangles lift to straight lines [13, 7, 3]. For $\mathfrak{b}(p, q)$ the two lifted lines have classes $(0, 1)$ and (p, q) ; they meet in $|\det \begin{pmatrix} 0 & 1 \\ p & q \end{pmatrix}| = p$ points, and gluing the two solid tori recovers $\Sigma_2(K) = L(p, q)$. Being straight lines, they bound no immersed bigon.

Two things follow, and one warning. First, the naive count on P itself is $(p+1)/2$: the $(p-1)/2$ interior dihedral characters of Theorem 1.1 together with the single abelian character at a corner. The passage to the full $\det = p$ requires the reduction defining $I^\natural$, i.e. the earring. Second, and structurally: the differential and the bounding-cochain deformation are assembled from holomorphic bigons and discs, of which straight lines have none. Thus on the two-bridge family the figure-eight bubbling that obstructs the general conjecture is inert.

This is consistent with the known outcome: Hedden–Herald–Kirk [7, 8] prove that for every two-bridge knot the $\mathbb{Z}/4$ gradings agree and the differentials vanish on both sides, so pillowcase homology is isomorphic to $I^\natural$ with zero bounding cochain, of rank $\det$. We emphasize this is *their* theorem; Theorem 1.1 only clarifies why the interior generators are exactly the dihedral characters and why the obstruction is corner-localized.

Remark 3.2 (A correction). It is tempting to hope that composing the straight Lagrangians with the Cazassus–Herald–Kirk–Kotelskiy earring correspondence $E \subset P^- \times P$ [3] preserves straightness. It does not: Herald–Kirk [9] prove, at the regular-homotopy level, that E sends an arc-type Lagrangian to a genuine figure-eight curve with one self-crossing per corner, while doubling loop-type curves that avoid the corners. Thus $E \circ L_1$ is immersed, and all of its new

geometry is confined to the four corners. For two-bridge knots the net effect is nonetheless trivial (zero bounding cochain suffices), and every known example with nonzero bounding cochain—the $(4, 5)$ -torus knot [3] and the pretzel knot $P(-2, 3, 5)$ [16]—is non-alternating.

4. THE $(3, n)$ -TORUS KNOTS

4.1. The representation arcs. Write $\pi_1(S^3 \setminus T(3, n)) = \langle x, y \mid x^3 = y^n \rangle$ with $\gcd(3, n) = 1$. The element $z = x^3 = y^n$ is central, so any irreducible $\mathrm{SU}(2)$ representation sends $z \mapsto \pm 1$ and

$$\rho(x) = e^{\alpha \hat{u}}, \quad \rho(y) = e^{\beta \hat{v}}, \quad \alpha = \frac{\pi \ell_1}{3}, \quad \beta = \frac{\pi \ell_2}{n},$$

with $\ell_1 \in \{1, 2\}$, $0 < \ell_2 < n$, and the central-sign match $\ell_1 \equiv \ell_2 \pmod{2}$. The angle $\theta = \angle(\hat{u}, \hat{v}) \in (0, \pi)$ is the only remaining modulus; each admissible pair (ℓ_1, ℓ_2) is a one-parameter arc of irreducible characters (Klassen [10]).

4.2. The trace-free locus. The meridian is $\mu = x^a y^b$ with $an + 3b = 1$; it is well defined modulo the center, and trace-freeness is insensitive to that ambiguity (multiplying μ by z negates $\rho(\mu)$, preserving trace 0). A direct quaternion computation gives

$$\frac{1}{2} \mathrm{tr} \rho(\mu) = \mathrm{Re} \rho(\mu) = \cos(a\alpha) \cos(b\beta) - \sin(a\alpha) \sin(b\beta) \cos \theta,$$

which is affine in $\cos \theta$. Hence each arc contains exactly one trace-free character, at

$$\cos \theta = \cot(a\alpha) \cot(b\beta), \tag{1}$$

provided the right side lies in $[-1, 1]$ (otherwise the arc carries none). We verified in each case that the resulting ρ satisfies $\rho(x)^3 = \rho(y)^n = -1$ and $\mathrm{tr} \rho(\mu) = 0$ exactly.

Proposition 4.1 (Count). *The number of irreducible trace-free $\mathrm{SU}(2)$ characters of $T(3, n)$ is*

$$N(3, n) = \begin{cases} 2(n-1)/3, & n \equiv 1, \\ (2n-1)/3, & n \equiv 2, \\ (2n+1)/3, & n \equiv 4, \\ 2(n+1)/3, & n \equiv 5, \end{cases} \pmod{6},$$

together with one abelian trace-free character; thus $N(3, n) \sim 2n/3$.

The values $N(3, n)$ for $n = 2, 4, 5, 7, 8, 10, 11, 13$ are $1, 3, 4, 4, 5, 7, 8, 8$. These count the admissible arcs, i.e. those on which the right side of (1) lies in $[-1, 1]$; the formula was checked by direct enumeration of the admissible arcs for all $n \leq 25$.

4.3. The dihedral dichotomy.

Proof of Theorem 1.2. In $\mathrm{Pin}(2) = S^1 \cup jS^1$ every element is a rotation (in S^1) or a reflection (in jS^1); reflections are exactly the trace-free elements, and an irreducible dihedral representation must send a generator to a reflection. Now $\rho(x)$ has trace $2 \cos(\pi \ell_1/3) = \pm 1 \neq 0$, so it is never trace-free (as 3 is odd); and $\rho(y)$ is trace-free iff $\beta = \pi/2$ iff $\ell_2 = n/2$, possible only for n even. Hence the number of irreducible dihedral trace-free characters is 1 if n is even and 0 if n is odd, which equals $(\det T(3, n) - 1)/2$ since $\det T(3, n) = 3$ for n even and 1 for n odd. In particular, for n odd there are no dihedral characters, so every irreducible trace-free character is non-dihedral. $\square$

Theorem 1.2 is the exact opposite of Theorem 1.1: two-bridge knots are entirely dihedral, the $(3, n)$ -torus knots with n odd entirely non-dihedral. The non-dihedral characters are precisely those the pillowcase construction represents by non-straight Lagrangians, on which the earring produces genuine figure-eight curves; this is why the family is a natural first test beyond the metabelian world.

Remark 4.2. There is a visible symmetry: $(\ell_1, \ell_2) \mapsto (3 - \ell_1, n - \ell_2)$ preserves θ in (1), pairing the two central-sign sheets $\rho(x)^3 = \pm 1$ at equal meridian angle.

5. THE $\mathbb{Z}/4$ GRADINGS VIA THE DOUBLE BRANCHED COVER

The double branched cover of $T(3, n)$ is the Brieskorn sphere $\Sigma(2, 3, n)$. For n odd it is an integral homology sphere; its irreducible flat $\mathrm{SO}(3)$ connections have rotation numbers $(m_1, m_2, m_3) = (1, m_2, m_3)$ indexing the three singular fibers, with m_2, m_3 even, $0 < m_2 < 3$, $0 < m_3 < n$. Since 2 is the only even value in $(0, 3)$, necessarily $m_2 = 2$, and m_3 ranges over the even integers in $(0, n)$ meeting the spherical-triangle inequalities on the angles $(\pi/2, 2\pi/3, \pi m_3/n)$. (These m_i index the branched cover and are not the (ℓ_1, ℓ_2) of Section 4.) Their number is $a = -\sigma(T(3, n))/4$, and each lifts to two trace-free knot characters, so

$$N(3, n) = 2a \quad (n \text{ odd}), \quad \sigma(T(3, n)) = -2N(3, n), \quad (2)$$

which we verified (independent triangle-inequality count) for all odd $n \leq 43$. Thus the character count of Proposition 4.1 determines the signature and, with the grading formulae below, the graded homology.

5.1. The grading formula. The $\mathbb{Z}/4$ grading is the spectral flow of the extended Hessian mod 4, relative to the trivial connection Θ (a single generator, in grading $\sigma \bmod 4$); every irreducible contributes four generators, two in grading μ and two in $\mu + 1$ [12, 14]. The knot grading is

$$\mu(\alpha) = \frac{1}{2} \mathrm{gr}(\alpha) + \frac{1}{4}(1 - \rho_{\mathrm{Ad}} \alpha) \pmod{4},$$

where, for $\Sigma(2, p, q)$ with $p = 3$, $q = n$, the Fintushel–Stern index [6, 1] and the equivariant ρ -invariant [1] are

$$\mathrm{gr}(\alpha) = \frac{e^2}{pq} + \frac{2}{p} \sum_{k=1}^{p-1} \cot \frac{\pi k}{p} \cot \frac{\pi b_2 k}{p} \sin^2 \frac{\pi k m_2}{p} + \frac{2}{q} \sum_{k=1}^{q-1} \cot \frac{\pi k}{q} \cot \frac{\pi b_3 k}{q} \sin^2 \frac{\pi k m_3}{q}, \quad (3)$$

$$\rho_{\mathrm{Ad}} \alpha = 1 - \frac{2}{pq} - \frac{4}{p} \sum_{k=1}^{p-1} \cot \frac{\pi k}{p} \tan \frac{\pi b_2 k}{p} \cos^2 \frac{\pi k m_2}{p} - \frac{4}{q} \sum_{k=1}^{q-1} \cot \frac{\pi k}{q} \tan \frac{\pi b_3 k}{q} \cos^2 \frac{\pi k m_3}{q}, \quad (4)$$

with $e = pqm_1 + 2qm_2 + 2pm_3$ and b_2, b_3 the even Seifert invariants ($pqb_1 + 2qb_2 + 2pb_3 = 1$); since the cotangents see them only mod p, q , one may take $b_2 \equiv (2q)^{-1}$, $b_3 \equiv (2p)^{-1}$.

Example 5.1. For $\Sigma(2, 3, 7)$ the two connections $(1, 2, 2)$ and $(1, 2, 4)$ give $\mathrm{gr} = 175 \equiv 7$ and $\mathrm{gr} = 251 \equiv 3 \pmod{8}$, $\rho = 3$ in both cases, and $\mu = 87 \equiv 3$ and $\mu = 125 \equiv 1 \pmod{4}$. This reproduces Anvari's Example 6.1 [1] to the raw values, and calibrates our implementation.

5.2. The gradings of $T(3, n)$, n odd. Evaluating (3)–(4) over the admissible connections gives the following.

Proposition 5.2. *Let n be odd and coprime to 3, and set $a = -\sigma(T(3, n))/4 = N(3, n)/2$ (an even integer). Under the grading (3) the a irreducible connections of $\Sigma(2, 3, n)$ split evenly: $a/2$ have $\mu = 1$ and $a/2$ have $\mu = 3$ (never 0 or 2). Consequently the reduced singular-instanton chain complex is*

$$IC^\natural(T(3, n)) = (1 + a, a, a, a), \quad \chi = +1,$$

with the trivial generator Θ in grading 0. This is the Poudel–Saveliev chain-level statement [14, §7.4] (Anvari [1] computes exactly these generators and gradings; the irreducible part is Daemi–Scaduto's [4]). The homology $I^\natural$ then splits by $n \bmod 6$:

$$\mathrm{rank} I^\natural(T(3, n)) = \sum_i |\Delta_{T(3, n)}| = \begin{cases} 1 + 4a, & n \equiv 1 \pmod{6} \quad (\partial = 0), \\ 4a - 1 = (1 + 4a) - 2, & n \equiv 5 \pmod{6} \quad (\mathrm{rank} \partial = 1). \end{cases}$$

For $n \equiv 1$ the Alexander lower bound $\sum_i |\Delta| = 1 + 4a$ meets the chain rank, forcing $\partial = 0$ and $I^\natural = IC^\natural = (1 + a, a, a, a)$ (Anvari [1, Ex. 6.2] proves this family). For $n \equiv 5$ the differential is nonzero; the smallest case is $T(3, 5) = P(-2, 3, 5) = 10_{124}$, where $\text{rank } I^\natural = 7$ (not 9), pinned by $\ell = u = 7$ [16], and where on the pillowcase side the nonzero bounding cochain of Smith [16] enacts the cancellation.

Proof and verification. On $\Sigma(2, 3, n)$ one has $\rho \equiv 3$, so $\mu = (\text{gr} - 1)/2$. A $\mu = 3$ connection contributes two chain generators to each of gradings 3 and 0; a $\mu = 1$ connection two to each of 1 and 2. That a is even follows from $a = (n - 1)/3 = 2m$ for $n = 6m + 1$ and $a = (n + 1)/3 = 2m + 2$ for $n = 6m + 5$; so with $a/2$ of each type the irreducibles give (a, a, a, a) , and Θ adds one to grading 0, giving $IC^\natural = (1 + a, a, a, a)$ of chain rank $1 + 4a$. The even split (hence this chain complex) is proved for $n \equiv 1 \pmod{6}$ by Anvari [1, Ex. 6.2] and verified here by direct evaluation of (3)–(4) for all odd $n \leq 43$. For the homology, $\sum_i |\Delta_{T(3, n)}| = 1 + 4a$ when $n \equiv 1$ and $4a - 1$ when $n \equiv 5$; since the Khovanov–Alexander bounds give $\sum_i |\Delta| \leq \text{rank } I^\natural \leq \text{rank } \widetilde{Kh}$ and, for these knots, $\text{rank } \widetilde{Kh} = \sum_i |\Delta|$, the differential vanishes iff $n \equiv 1$; for $n \equiv 5$ it has rank 1. (The case $T(3, 5)$ is Smith’s [16], rigorous with $\ell = u = 7$.) $\square$

Explicitly $a = (n - 1)/3$ for $n \equiv 1$ and $a = (n + 1)/3$ for $n \equiv 5 \pmod{6}$; either way $\text{rank } I^\natural \sim 2N(3, n)$ grows while $\det = 1$ is fixed, so $I^\natural(T(3, n))$ is far from thin. We note that $\chi(I^\natural(K)) = +1$ for *every* knot [14, Thm 8.1]—carried by the single trivial generator—so here χ coincides with $\det = 1$ only by accident; in general $\chi \neq \det$, as the non-thin case 8_{19} below ($\chi = 1$, $\det = 3$) shows.

n	5	7	11	13	17	19	23	25
$n \bmod 6$	5	1	5	1	5	1	5	1
$N(3, n) = 2a$	4	4	8	8	12	12	16	16
$a = -\sigma/4$	2	2	4	4	6	6	8	8
chain rank $1 + 4a$	9	9	17	17	25	25	33	33
rank $I^\natural$	7	9	15	17	23	25	31	33

TABLE 1. For $T(3, n)$, n odd coprime to 3: the trace-free character count $N = 2a$, the chain rank $\dim IC^\natural = 1 + 4a$ (Proposition 5.2), and the homology rank $\text{rank } I^\natural = \sum_i |\Delta_{T(3, n)}|$. The two coincide for $n \equiv 1 \pmod{6}$ (differential zero) but differ by 2 for $n \equiv 5 \pmod{6}$ (bold; differential rank 1). $T(3, 5) = P(-2, 3, 5) = 10_{124}$ is rigorous ($\ell = u = 7$, [16]); the graded ranks for $n \equiv 1$ are $(1 + a, a, a, a)$.

The Fintushel–Stern grading is special to n odd, where $\Sigma(2, 3, n)$ is a homology sphere; for n even $b_3 \equiv (2p)^{-1} \pmod{n}$ does not exist. A nonzero differential appears both for $n \equiv 5 \pmod{6}$ (above) and for n even. The smallest and cleanest instance—bounding cochain vanishing, differential a single bigon—is $8_{19} = T(3, 4)$, to which we now turn.

6. A NONZERO DIFFERENTIAL: $8_{19} = T(3, 4)$

For n even the double cover $\Sigma(2, 3, n)$ carries nontrivial reducibles ($H_1 \neq 0$), and the pillowcase complex acquires a differential. We treat the smallest case, $8_{19} = T(3, 4)$, in two ways.

6.1. Structural derivation. By Theorem 1.2 and Proposition 4.1, $T(3, 4)$ has three irreducible trace-free characters—two non-dihedral (at $\theta \approx 125.3^\circ, 54.7^\circ$) and one dihedral (at 90°)—together with one abelian character. Via the double cover $\Sigma(2, 3, 4)$, whose first homology is $\mathbb{Z}/3$: the two non-dihedral characters are the two lifts of the single Σ -irreducible; the dihedral character is the

nontrivial reducible; the abelian character is Θ . Through the grading framework (irreducible $\rightarrow$ four generators at $(\mu, \mu, \mu+1, \mu+1)$, nontrivial reducible $\rightarrow$ two, $\Theta \rightarrow$ one at $\sigma \bmod 4 = 2$), with $\mu_{\text{irr}} = 3$ and $\mu_{\text{red}} = 1$, the chain complex assembles to

$$C^{\natural} = (2, 1, 2, 2), \quad \chi = +1,$$

matching the instanton computation of Poudel–Saveliev [14, Ex. 7.9]. A finite search over grading-respecting differentials shows that the *only* differential compatible with homology $(2, 1, 1, 1)$ is a single rank-one map $\partial_3: C_3 \rightarrow C_2$: one bigon from a grading-3 generator (a lift of the irreducible, an interior point) to a grading-2 generator (the reducible/ Θ , a corner point). We derived this before consulting [8]; it predicts an interior-to-corner bigon.

6.2. The explicit bigon. Hedden–Herald–Kirk [8, §11.6] realize exactly this. With L_0 the earring figure-eight and L_1 the perturbed character variety of the torus-knot tangle $(p, q, r, s) = (3, 4, 3, -2)$, the singular unperturbed variety (a “ φ -curve”) resolves into an arc R_0 and a vertically monotone circle R_1 . The seven generators are r_+ (corner, $\text{gr} = 2$), $x_1^{\pm}$ (arc), and $x_2^{\pm}, x_3^{\pm}$ (circle); the gradings are $\text{gr}(x_1^+) = 0$, $\text{gr}(x_1^-) = 3$, and the circle contributes one generator in each grading. There is exactly one bigon, $x_1^- \rightarrow r_+$, running into the corner along R_0 and the earring, giving $\partial x_1^- = r_+$ —a rank-one map from grading 3 to grading 2. The circle bounds no bigon. The homology is

$$H^{\natural} = (1, 0, 0, 0)_{\text{arc}} \oplus (1, 1, 1, 1)_{\text{circle}} = (2, 1, 1, 1) = I^{\natural}(8_{19}),$$

of rank $5 > \det = 3$; the complex is non-thin.

6.3. The corner-bigon thread. The structural derivation and the explicit computation give the same graded complex and the same differential. The bigon is a corner figure-eight bigon—the earring’s self-crossing at a pillowcase corner linking an interior irreducible character to the reducible locus. This is precisely the mechanism absent on two-bridge knots (Section 3.2): there the Lagrangians are straight, there is no corner bigon, and $\partial = 0$; here the single corner bigon is what makes $I^{\natural}(8_{19})$ non-thin. The geometric computation is due to [8]; our contribution is the independent structural derivation predicting the differential’s rank, source, and target from the character count alone.

7. THE OPEN DIRECTION

Everything above concerns generators and gradings; the analytic heart of the knot Atiyah–Floer conjecture is the bounding cochain. On two-bridge knots it is inert (Section 3.2); off them it is genuinely nonzero, as shown for the $(4, 5)$ -torus knot [3] and the pretzel knot $P(-2, 3, 5)$ [16]. Notably $P(-2, 3, 5) = T(3, 5) = 10_{124}$ is itself a member of our family—the smallest $n \equiv 5 \pmod{6}$ case—so the nonzero differential of Proposition 5.2 for $n \equiv 5$ is, on the pillowcase side, precisely Smith’s nonzero bounding cochain. A differential thus appears for $n \equiv 5$ (odd) and for all n even; two problems, of increasing difficulty, remain.

First, the instanton side. For every torus knot the irreducible singular instanton homology is $\mathbb{Z}_{(1)}^{[-\sigma/4]} \oplus \mathbb{Z}_{(3)}^{[-\sigma/4]}$ with vanishing differential [4]. For n even, however, $\Sigma(2, 3, n)$ carries a nontrivial reducible ($\det = 3$, so $H_1 = \mathbb{Z}/3$), and the reduced complex has the shape

$$C^{\natural}(T(3, n)) = \underbrace{\frac{N-1}{2} \text{ irreducibles}}_{4 \text{ generators each}} \oplus \underbrace{\text{one reducible}}_2 \oplus \Theta, \quad \dim = 2N(3, n) + 1,$$

on which an irreducible-to-reducible differential can appear—exactly the corner figure-eight bigon of 8_{19} (Section 6). Organizing this differential across the even family—in particular, whether it is always of rank one, as for 8_{19} , giving $I^{\natural}$ of rank $2N(3, n) - 1$ —is open beyond the single case $n = 4$ [14].

Second, and harder, the pillowcase side: to compute the bounding-cochain-deformed pillowcase invariant and match it to $I^{\natural}$. No bounding cochain has been computed for an infinite family in any setting; the record is the $(4, 5)$ -torus knot (correct rank [3]) and $P(-2, 3, 5)$ (nonzero only [16]). The immersed-curve machinery of [9] is the natural tool, but its assembly into a family computation is not yet in place. Both problems lie beyond the present, purely representation-theoretic, account.

REFERENCES

- [1] N. Anvari, *Equivariant rho-invariants and instanton homology of torus knots*, arXiv:1609.05025.
- [2] M. F. Atiyah, *New invariants of 3- and 4-dimensional manifolds*, in *The mathematical heritage of Hermann Weyl*, Proc. Sympos. Pure Math. **48** (1988), 285–299.
- [3] G. Cazassus, C. Herald, P. Kirk, A. Kotelskiy, *The correspondence induced on the pillowcase by the earring tangle*, J. Topol. **15** (2022), 2472–2543. arXiv:2010.04320.
- [4] A. Daemi, C. Scaduto, *Chern–Simons functional, singular instantons, and the four-dimensional clasp number*, J. Eur. Math. Soc. **26** (2024), 2127–2190. arXiv:2007.13160.
- [5] S. Dostoglou, D. Salamon, *Self-dual instantons and holomorphic curves*, Ann. of Math. **139** (1994), 581–640.
- [6] R. Fintushel, R. Stern, *Instanton homology of Seifert fibred homology three spheres*, Proc. London Math. Soc. **61** (1990), 109–137.
- [7] M. Hedden, C. Herald, P. Kirk, *The pillowcase and perturbations of traceless representations of knot groups*, Geom. Topol. **18** (2014), 211–287. arXiv:1301.0164.
- [8] M. Hedden, C. Herald, P. Kirk, *The pillowcase and traceless representations of knot groups II: a Lagrangian–Floer theory in the pillowcase*, J. Symplectic Geom. **16** (2018), 721–815. arXiv:1501.00028.
- [9] C. Herald, P. Kirk, *An endomorphism on immersed curves in the pillowcase*, arXiv:2407.11247.
- [10] E. Klassen, *Representations of knot groups in $SU(2)$* , Trans. Amer. Math. Soc. **326** (1991), 795–828.
- [11] P. Kronheimer, T. Mrowka, *Knot homology groups from instantons*, J. Topol. **4** (2011), 835–918. arXiv:0806.1053.
- [12] P. Kronheimer, T. Mrowka, *Khovanov homology is an unknot-detector*, Publ. Math. IHÉS **113** (2011), 97–208. arXiv:1005.4346.
- [13] X.-S. Lin, *A knot invariant via representation spaces*, J. Differential Geom. **35** (1992), 337–357.
- [14] P. Poudel, N. Saveliev, *Link homology and equivariant gauge theory*, Algebr. Geom. Topol. **17** (2017), 2635–2685. arXiv:1502.03116.
- [15] R. Riley, *Nonabelian representations of 2-bridge knot groups*, Quart. J. Math. Oxford **35** (1984), 191–208.
- [16] K. Smith, *Perturbed traceless $SU(2)$ character varieties of tangle sums*, arXiv:2412.06066.

BERND J. WUEBBEN, NEW YORK, NY, wuebben@gmail.com